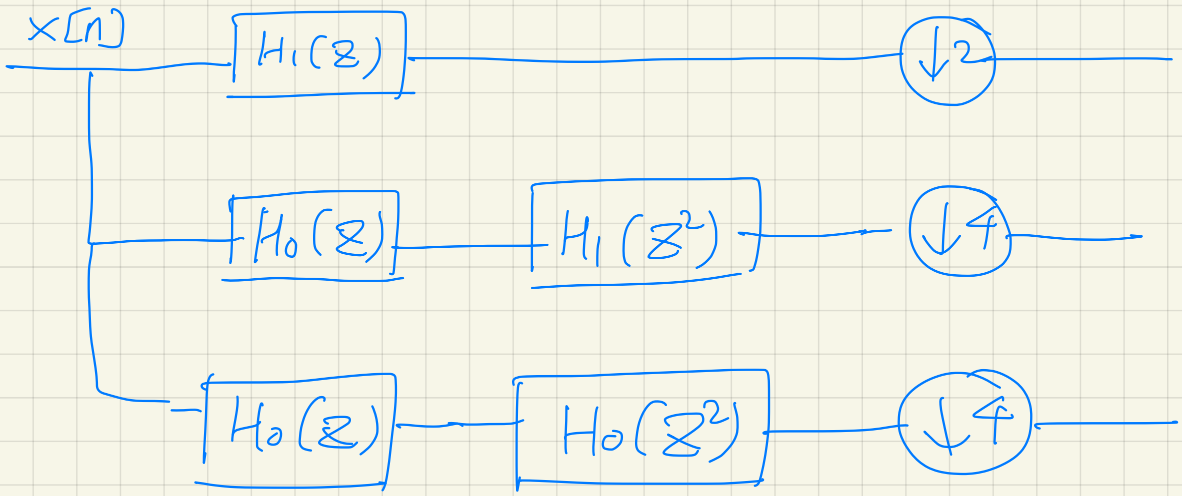
Equivalent system?



$$s = 1 - z^{-1}$$

$$H(z) = H_c(s)\Big|_{s=1-z^{-1}}$$

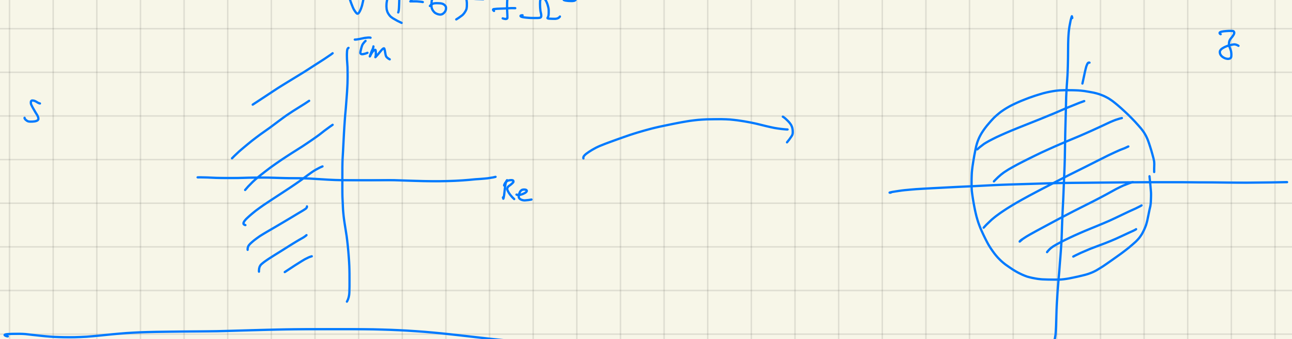
Q1) stable $H_c(s) \Rightarrow$ stable $H(z)$?

Q2) stable $H(z) \Rightarrow$ stable $H_c(s)$?

$s = \sigma + j\Omega \qquad \sigma < 0$

$s = 1 - z^{-1} \qquad z^{-1} = 1 - s \qquad z = \frac{1}{1-s} = \frac{1}{1-\sigma-j\Omega}$

$$|z| = \frac{1}{\sqrt{(1-\sigma)^2 + \Omega^2}} < 1$$



$z = re^{j\omega} \qquad 0 < r < 1$

$$s = 1 - z^{-1} = 1 - r^{-1}e^{-j\omega}$$

$$= \underbrace{1 - r^{-1}\cos\omega} + j\underbrace{r^{-1}\sin\omega}$$

$Re(s) = 1 - r^{-1}\cos\omega$

$Re(s)$ is not always negative

$$x[n] \longrightarrow \boxed{\uparrow 3} \longrightarrow \underbrace{\boxed{\downarrow 2} \longrightarrow \boxed{\downarrow 3}}_{\downarrow 6} \longrightarrow \boxed{\uparrow 2} \longrightarrow y[n]$$

Q: Can we simplify?

Q1: $\longrightarrow \boxed{\uparrow L} \longrightarrow \boxed{\downarrow L} \longrightarrow$

Q2: $\xrightarrow{x[n]} \boxed{\uparrow M} \xrightarrow{y[n]} \boxed{\downarrow L} \xrightarrow{z[n]} \quad \equiv \quad \xrightarrow{x[n]} \boxed{\downarrow L} \xrightarrow{v[n]} \boxed{\uparrow M} \xrightarrow{w[n]}$

when?

If M and L are relatively prime.

$$Y(e^{j\omega}) = X(e^{jM\omega})$$

$$Z(e^{j\omega}) = \frac{1}{L}\sum_{k=0}^{L-1} Y\left(e^{j\frac{\omega-2\pi k}{L}}\right)$$

$$= \frac{1}{L}\sum_{k=0}^{L-1} X\left(e^{jM\frac{\omega-2\pi k}{L}}\right)$$

$$V(e^{j\omega}) = \frac{1}{L}\sum_{k=0}^{L-1} X\left(e^{j\frac{\omega-2\pi k}{L}}\right)$$

$$W(e^{j\omega}) = V(e^{jM\omega})$$

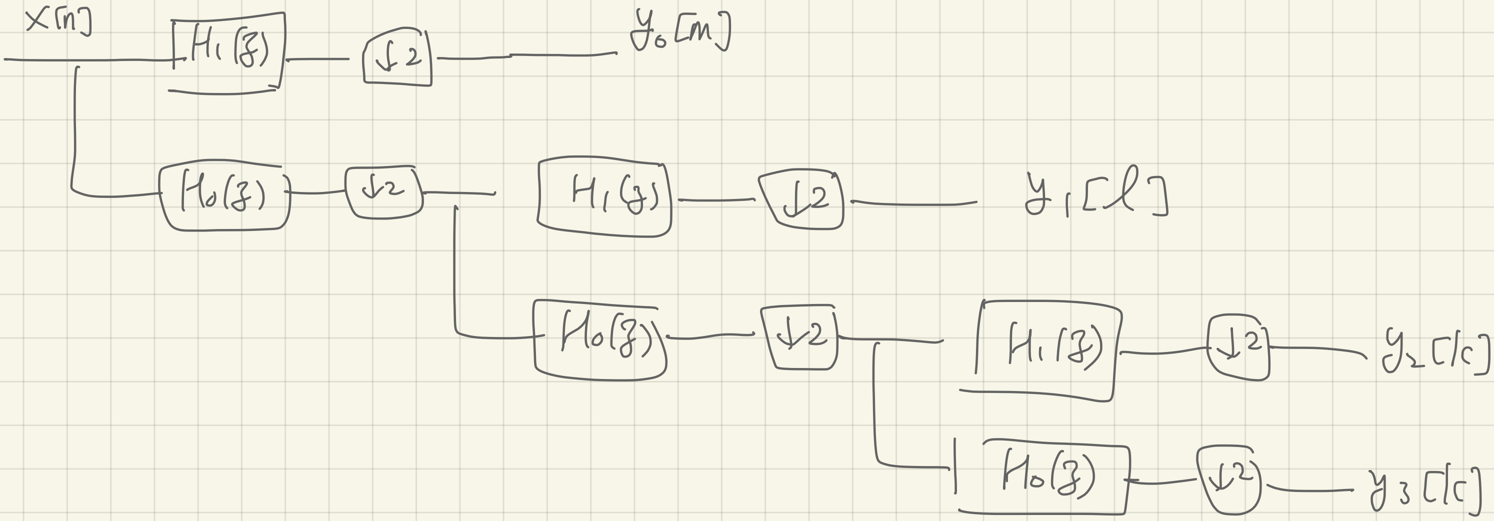
$$= \frac{1}{L}\sum_{k=0}^{L-1} X\left(e^{j\frac{M\omega-2\pi k}{L}}\right)$$

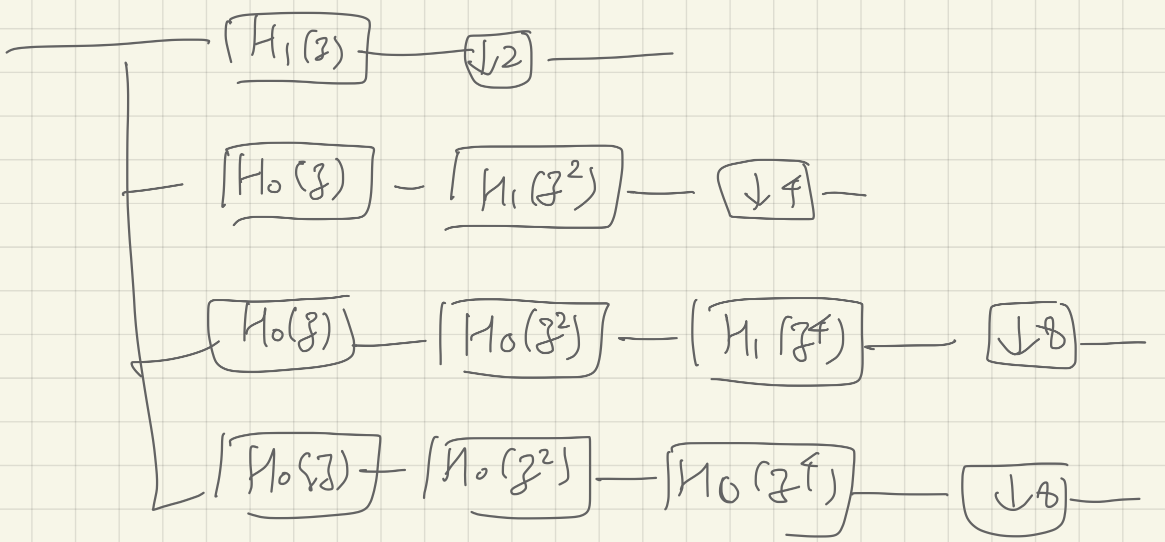
$$\frac{M\omega - 2\pi M k}{L}$$

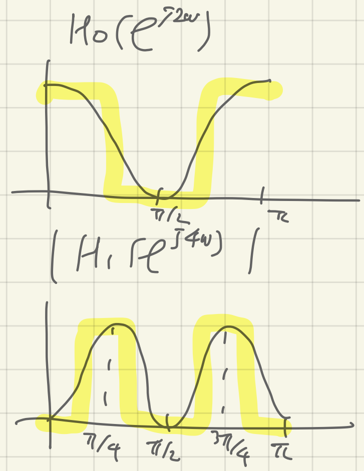
$$\frac{M\omega - 2\pi k}{L}$$

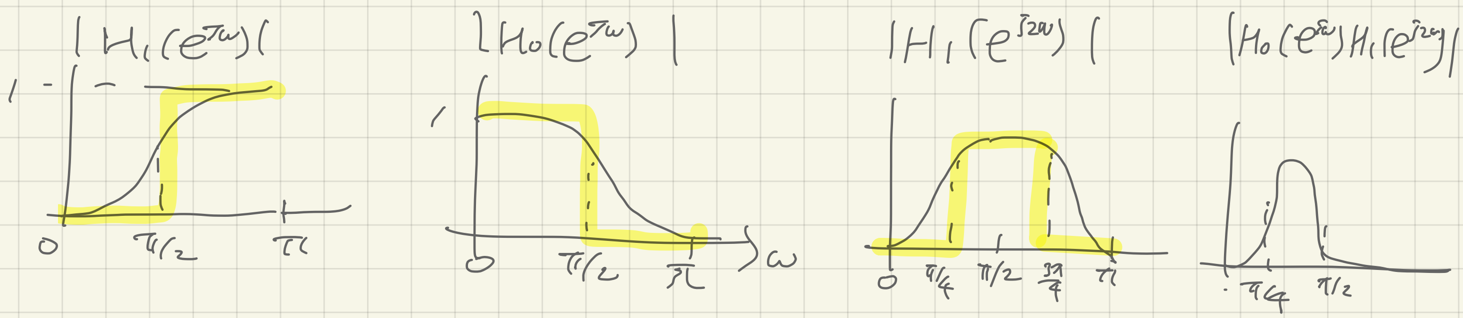
$L = 3 \quad M = 2$

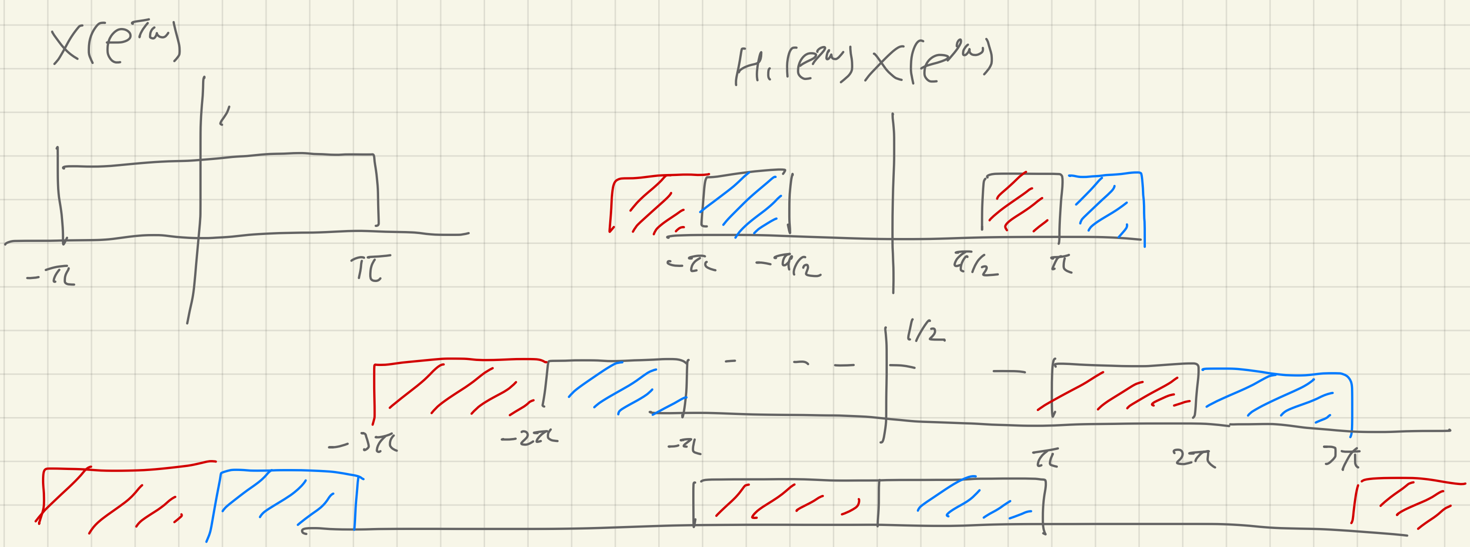
k	0	1	2
$\frac{2\omega - 4\pi k}{3}$	$\frac{2\omega}{3}$	$\frac{2\omega - 4\pi}{3}$	$\frac{2\omega - 8\pi}{3}$
$\frac{2\omega - 2\pi k}{3}$	$\frac{2\omega}{3}$	$\frac{2\omega - 2\pi}{3}$	$\frac{2\omega - 4\pi}{3}$

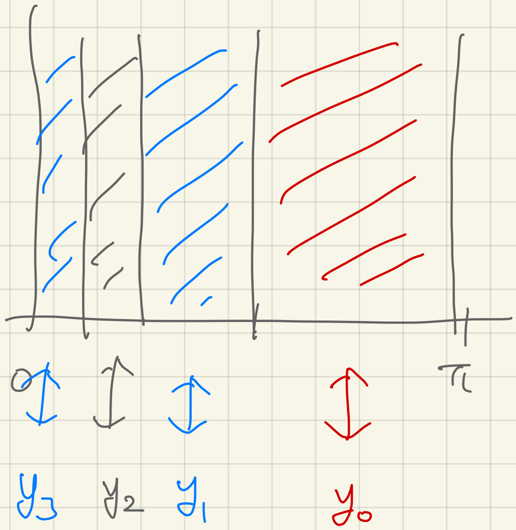

x[n]
H1(z)
↓2
y0[m]
H0(z)
↓2
H1(z)
↓2
y1[l]
H0(z)
↓2
H1(z)
↓2
y2[k]
H0(z)
↓2
y3[k]


H1(z)
↓2
H0(z)
H1(z²)
↓4
H0(z)
H0(z²)
H1(z⁴)
↓8
H0(z)
H0(z²)
H0(z⁴)
↓8

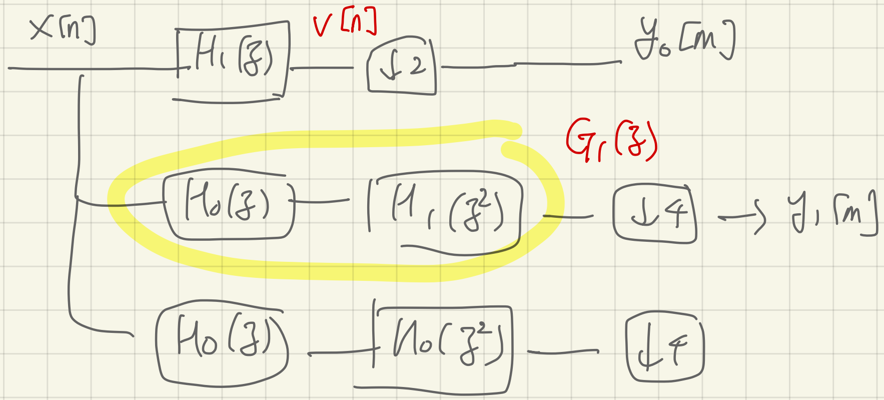

H0(e^j2ω)
π/2
π
|H1(e^j4ω)|
π/4
π/2
3π/4
π


|H1(e^jω)|
1
0
π/2
π
|H0(e^jω)|
1
0
π/2
π
ω
|H1(e^j2ω)|
0
π/4
π/2
3π/4
π
|H0(e^jω)H1(e^j2ω)|
π/4
π/2


X(e^jω)
1
−π
π
H1(e^jω)X(e^jω)
−π
−π/2
π/2
π
1/2
−3π
−2π
−π
π
2π
3π



0
π_1
y_3
y_2
y_1
y_0



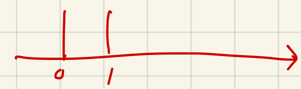
$$G_1(z) = H_0(z)\, H_1(z^2)$$

$$g_1[n] = h_0[n] * \left(\boxed{\uparrow 2}\, h_1[n]\right)$$

$$\tilde{h}_0[n] = \frac{\delta[n] + \delta[n-1]}{\sqrt{2}}$$

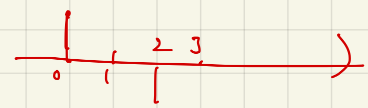
$$\tilde{h}_1[n] = \frac{\delta[n] - \delta[n-1]}{\sqrt{2}}$$

$\tilde{h}_0[n]$



$\tilde{h}_1[n] = h_1[-n]$: $h_1[n]$ is real-valued

$\boxed{\uparrow 2}\, \tilde{h}_1[n]$



$$v[n] = h_1[n] * x[n] = \sum_{k=-\infty}^{\infty} x[k]\, h_1[n-k]$$

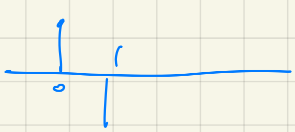
$$y_0[m] = v[2m] = \sum_{k=-\infty}^{\infty} x[k]\, h_1[2m-k]$$

$$= \underbrace{\sum_{k=-\infty}^{\infty} x[k]\, \tilde{h}_1[k-2m]}$$

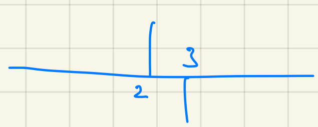
dot product of $x[n]$ and $\tilde{h}_1[n-2m]$

Example $\quad \tilde{h}[n] = \dfrac{\delta[n] - \delta[n-1]}{\sqrt{2}}$

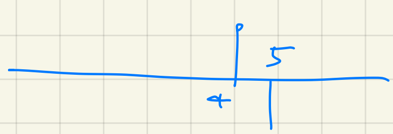
$\tilde{h}[n]$



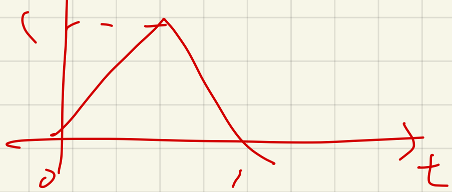
$\tilde{h}[n-2]$



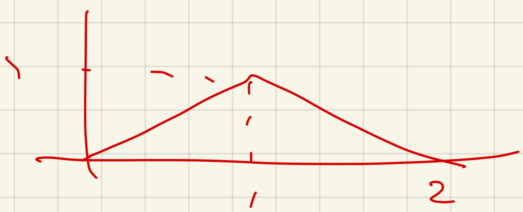
$\tilde{h}[n-4]$



$x(t)$



$x\left(\frac{t}{2}\right)$



$x(2t)$

$x[n]$